Experiment – 1

Aim: Design an application to perform the arithmetic operations by reading the input from user

Solution:

activity_main.xml:

```
<?xml version="1.0" encoding="utf-8"?>
<androidx.constraintlayout.widget.ConstraintLayout
xmlns:android="http://schemas.android.com/apk/res/android"
   xmlns:app="http://schemas.android.com/apk/res-auto"
   xmlns:tools="http://schemas.android.com/tools"
   android:layout_width="match_parent"
   android:layout_height="match_parent"
   tools:context=".MainActivity">

   <EditText
      android:id="@+id/aBox"
      android:layout_width="wrap_content"
      android:layout_height="wrap_content"
      android:ems="10"
      android:minHeight="48dp"
      android:minWidth="48dp"
      android:hint="Enter a Value"
      android:inputType="text"
      app:layout_constraintEnd_toEndOf="parent"
      app:layout_constraintStart_toStartOf="parent"
      app:layout_constraintTop_toTopOf="parent"
      app:layout_constraintBottom_toTopOf="@+id/bBox"/>

   <EditText
      android:id="@+id/bBox"
      android:layout_width="wrap_content"
      android:layout_height="wrap_content"
      android:ems="10"
      android:hint="Enter b Value"
      android:inputType="text"
      android:minWidth="48dp"
      android:minHeight="48dp"
      app:layout_constraintBottom_toTopOf="@+id/addBtn"
      app:layout_constraintEnd_toEndOf="parent"
      app:layout_constraintStart_toStartOf="parent"
      app:layout_constraintTop_toBottomOf="@+id/aBox" />

   <Button
      android:id="@+id/addBtn"
      android:layout_width="wrap_content"
      android:layout_height="wrap_content"
      android:text="+"
      app:layout_constraintStart_toStartOf="parent"
      app:layout_constraintEnd_toStartOf="@+id/subBtn"
      app:layout_constraintBottom_toTopOf="@+id/mulBtn"
```

```
        app:layout_constraintTop_toBottomOf="@+id/bBox"/>

    <Button
        android:id="@+id/subBtn"
        android:layout_width="wrap_content"
        android:layout_height="wrap_content"
        android:text="-"
        tools:layout_editor_absoluteX="230dp"
        tools:layout_editor_absoluteY="322dp"
        app:layout_constraintBottom_toTopOf="@+id/divBtn"
        app:layout_constraintStart_toEndOf="@+id/addBtn"
        app:layout_constraintEnd_toEndOf="parent"
        app:layout_constraintTop_toBottomOf="@+id/bBox"/>

    <Button
        android:id="@+id/mulBtn"
        android:layout_width="wrap_content"
        android:layout_height="wrap_content"
        android:text="*"
        app:layout_constraintStart_toStartOf="parent"
        app:layout_constraintTop_toBottomOf="@+id/addBtn"
        app:layout_constraintEnd_toStartOf="@+id/divBtn"
        app:layout_constraintBottom_toTopOf="@+id/result"/>

    <Button
        android:id="@+id/divBtn"
        android:layout_width="wrap_content"
        android:layout_height="wrap_content"
        android:text="/"
        tools:layout_editor_absoluteX="230dp"
        tools:layout_editor_absoluteY="322dp"
        app:layout_constraintBottom_toTopOf="@+id/result"
        app:layout_constraintStart_toEndOf="@+id/mulBtn"
        app:layout_constraintEnd_toEndOf="parent"
        app:layout_constraintTop_toBottomOf="@+id/subBtn" />

    <TextView
        android:id="@+id/result"
        android:layout_width="wrap_content"
        android:layout_height="wrap_content"
        app:layout_constraintBottom_toBottomOf="parent"
        app:layout_constraintEnd_toEndOf="parent"
        app:layout_constraintStart_toStartOf="parent"
        app:layout_constraintTop_toBottomOf="@+id/divBtn" />
</androidx.constraintlayout.widget.ConstraintLayout>
```

MainActivity.java:

```
package com.example.arith;

import androidx.appcompat.app.AppCompatActivity;

import android.os.Bundle;
import android.view.View;
import android.widget.Button;
import android.widget.EditText;
import android.widget.TextView;

public class MainActivity extends AppCompatActivity {
    EditText aval, bval;
    Button add, sub, mul, div;
    TextView res;
    Double a, b;
    @Override
    protected void onCreate(Bundle savedInstanceState) {
        super.onCreate(savedInstanceState);
        setContentView(R.layout.activity_main);
        aval = findViewById(R.id.aBox);
        bval = findViewById(R.id.bBox);
        add = findViewById(R.id.addBtn);
        sub = findViewById(R.id.subBtn);
        mul = findViewById(R.id.mulBtn);
        div = findViewById(R.id.divBtn);
        res = findViewById(R.id.result);
        add.setOnClickListener(new View.OnClickListener() {
            @Override
            public void onClick(View v) {
                a = Double.parseDouble(aval.getText().toString());
                b = Double.parseDouble(bval.getText().toString());
                res.setText("Result: "+(a+b));
            }
        });
        sub.setOnClickListener(new View.OnClickListener() {
            @Override
            public void onClick(View v) {
                a = Double.parseDouble(aval.getText().toString());
                b = Double.parseDouble(bval.getText().toString());
                res.setText("Result: "+(a-b));
            }
        });
        mul.setOnClickListener(new View.OnClickListener() {
            @Override
            public void onClick(View v) {
                a = Double.parseDouble(aval.getText().toString());
                b = Double.parseDouble(bval.getText().toString());
                res.setText("Result: "+(a*b));
```

```
        }
    });
    div.setOnClickListener(new View.OnClickListener() {
        @Override
        public void onClick(View v) {
            a = Double.parseDouble(aval.getText().toString());
            b = Double.parseDouble(bval.getText().toString());
            if(b!=0)
                res.setText("Result: "+(a/b));
            else
                res.setText("Division not possible");
        }
    });
  }
}
```

Output:

Experiment – 2

Aim: Design an application to calculate the body mass index by reading the input from user

Solution:

activity_main.xml:

```
<?xml version="1.0" encoding="utf-8"?>
<androidx.constraintlayout.widget.ConstraintLayout
xmlns:android="http://schemas.android.com/apk/res/android"
   xmlns:app="http://schemas.android.com/apk/res-auto"
   xmlns:tools="http://schemas.android.com/tools"
   android:layout_width="match_parent"
   android:layout_height="match_parent"
   tools:context=".MainActivity">

   <TextView
      android:id="@+id/header"
      android:layout_width="wrap_content"
      android:layout_height="wrap_content"
      android:text="BMI Calculator"
      android:textSize="48dp"
      android:textStyle="bold"
      app:layout_constraintEnd_toEndOf="parent"
      app:layout_constraintStart_toStartOf="parent"
      app:layout_constraintTop_toTopOf="parent"
      app:layout_constraintBottom_toTopOf="@+id/wtLbl" />

   <TextView
      android:id="@+id/wtLbl"
      android:layout_width="wrap_content"
      android:layout_height="wrap_content"
      android:text="Weight (in Kg)"
      android:textStyle="bold"
      tools:layout_editor_absoluteX="53dp"
      tools:layout_editor_absoluteY="178dp"
      app:layout_constraintStart_toStartOf="parent"
      app:layout_constraintEnd_toStartOf="@+id/wtBox"
      app:layout_constraintBottom_toTopOf="@+id/htLbl"
      app:layout_constraintTop_toBottomOf="@+id/header" />

   <EditText
      android:id="@+id/wtBox"
      android:layout_width="wrap_content"
      android:layout_height="wrap_content"
      android:ems="10"
      android:hint="Enter Weight"
      android:inputType="text"
      android:minHeight="48dp"
      tools:layout_editor_absoluteX="184dp"
      tools:layout_editor_absoluteY="164dp"
```

```
    app:layout_constraintBottom_toTopOf="@+id/htBox"
    app:layout_constraintStart_toEndOf="@+id/wtLbl"
    app:layout_constraintTop_toBottomOf="@+id/header"
    app:layout_constraintEnd_toEndOf="parent"/>

<TextView
    android:id="@+id/htLbl"
    android:layout_width="wrap_content"
    android:layout_height="wrap_content"
    android:text="Height (in Cm)"
    android:textStyle="bold"
    tools:layout_editor_absoluteX="56dp"
    tools:layout_editor_absoluteY="295dp"
    app:layout_constraintTop_toBottomOf="@+id/wtLbl"
    app:layout_constraintStart_toStartOf="parent"
    app:layout_constraintEnd_toStartOf="@+id/wtBox"
    app:layout_constraintBottom_toTopOf="@+id/calcBtn"/>

<EditText
    android:id="@+id/htBox"
    android:layout_width="wrap_content"
    android:layout_height="wrap_content"
    android:ems="10"
    android:minHeight="48dp"
    android:hint="Enter Height"
    android:inputType="text"
    tools:layout_editor_absoluteX="184dp"
    tools:layout_editor_absoluteY="275dp"
    app:layout_constraintBottom_toTopOf="@+id/calcBtn"
    app:layout_constraintStart_toEndOf="@+id/htLbl"
    app:layout_constraintEnd_toEndOf="parent"
    app:layout_constraintTop_toBottomOf="@+id/wtBox"/>

<Button
    android:id="@+id/calcBtn"
    android:layout_width="wrap_content"
    android:layout_height="wrap_content"
    android:text="Calculate"
    tools:layout_editor_absoluteX="157dp"
    tools:layout_editor_absoluteY="431dp"
    app:layout_constraintEnd_toEndOf="parent"
    app:layout_constraintStart_toStartOf="parent"
    app:layout_constraintTop_toBottomOf="@id/htBox"
    app:layout_constraintBottom_toTopOf="@+id/result"/>

<TextView
    android:id="@+id/result"
    android:layout_width="wrap_content"
    android:layout_height="wrap_content"
    android:gravity="center_horizontal"
```

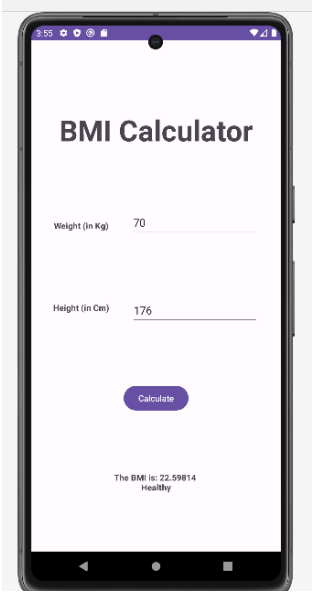
```
        android:textStyle="bold"
        tools:layout_editor_absoluteX="180dp"
        tools:layout_editor_absoluteY="571dp"
        app:layout_constraintEnd_toEndOf="parent"
        app:layout_constraintStart_toStartOf="parent"
        app:layout_constraintBottom_toBottomOf="parent"
        app:layout_constraintTop_toBottomOf="@+id/calcBtn"/>
</androidx.constraintlayout.widget.ConstraintLayout>
```

MainActivity.java:

```
package com.example.bmicalc;

import androidx.appcompat.app.AppCompatActivity;

import android.os.Bundle;
import android.view.View;
import android.widget.Button;
import android.widget.EditText;
import android.widget.TextView;
import android.widget.Toast;

public class MainActivity extends AppCompatActivity {
    EditText wtval, htval;
    Button btn;
    TextView res;

    @Override
    protected void onCreate(Bundle savedInstanceState) {
        super.onCreate(savedInstanceState);
        setContentView(R.layout.activity_main);
        wtval = findViewById(R.id.wtBox);
        htval = findViewById(R.id.htBox);
        btn = findViewById(R.id.calcBtn);
        res = findViewById(R.id.result);
        btn.setOnClickListener(new View.OnClickListener() {
            @Override
            public void onClick(View v) {
                Float wt = Float.parseFloat(wtval.getText().toString());
                Float ht_m = Float.parseFloat(htval.getText().toString())/100;
                Float bmi = wt/ (ht_m*ht_m);
                if(bmi < 18.5)
                    res.setText("The BMI is: "+bmi+" \n Under Weight ");
                else if (bmi >= 18.5 && bmi < 24.9)
                    res.setText("The BMI is: "+bmi+" \n Healthy ");
                else if (bmi >= 24.9 && bmi < 30)
                    res.setText("The BMI is: "+bmi+" \n Over Weight ");
                else if (bmi>=30)
                    res.setText("The BMI is: "+bmi+" \n Suffering from obesity ");
            }
        });
```

```
    }
}
```

Output:


BMI Calculator
Weight (In Kg)
70
Height (in Cm)
176
Calculate
The BMI is: 22.59814
Healthy

Experiment – 3

Aim: Design an application to fill the registration form using AutoCompleteTextView and RadioButton by reading the input from user

Solution:

activity_main.xml:

```
<?xml version="1.0" encoding="utf-8"?>
<androidx.constraintlayout.widget.ConstraintLayout
xmlns:android="http://schemas.android.com/apk/res/android"
   xmlns:app="http://schemas.android.com/apk/res-auto"
   xmlns:tools="http://schemas.android.com/tools"
   android:layout_width="match_parent"
   android:layout_height="match_parent"
   tools:context=".MainActivity">

   <TextView
      android:id="@+id/header"
      android:layout_width="wrap_content"
      android:layout_height="wrap_content"
      android:text="Registration Form"
      android:textSize="34sp"
      android:textStyle="bold"
      app:layout_constraintBottom_toTopOf="@+id/nameLbl"
      app:layout_constraintEnd_toEndOf="parent"
      app:layout_constraintHorizontal_bias="0.605"
      app:layout_constraintStart_toStartOf="parent"
      app:layout_constraintTop_toTopOf="parent" />

   <TextView
      android:id="@+id/nameLbl"
      android:layout_width="wrap_content"
      android:layout_height="wrap_content"
      android:text="Name"
      android:textStyle="bold"
      app:layout_constraintBottom_toTopOf="@+id/FNameLbl"
      app:layout_constraintEnd_toStartOf="@+id/NameBox"
      app:layout_constraintStart_toStartOf="parent"
      app:layout_constraintTop_toBottomOf="@+id/header" />

   <AutoCompleteTextView
      android:id="@+id/PlaceBox"
      android:layout_width="wrap_content"
      android:layout_height="wrap_content"
      android:hint="Enter Location"
      android:minHeight="48dp"
      app:layout_constraintBottom_toTopOf="@+id/SubmitBtn"
      app:layout_constraintStart_toEndOf="@+id/PlaceLbl"
      app:layout_constraintTop_toBottomOf="@+id/GenderGroup"
      app:layout_constraintEnd_toEndOf="parent"/>
```

```
<TextView
   android:id="@+id/PlaceLbl"
   android:layout_width="wrap_content"
   android:layout_height="wrap_content"
   android:text="Place of Birth"
   android:textStyle="bold"
   app:layout_constraintStart_toStartOf="parent"
   app:layout_constraintTop_toBottomOf="@+id/GenderGroup"
   app:layout_constraintEnd_toStartOf="@+id/PlaceBox"
   app:layout_constraintBottom_toTopOf="@+id/SubmitBtn"/>

<TextView
   android:id="@+id/GenderLbl"
   android:layout_width="wrap_content"
   android:layout_height="wrap_content"
   android:text="Gender"
   android:textStyle="bold"
   app:layout_constraintBottom_toTopOf="@+id/PlaceLbl"
   app:layout_constraintEnd_toStartOf="@+id/GenderGroup"
   app:layout_constraintStart_toStartOf="parent"
   app:layout_constraintTop_toBottomOf="@+id/FNameLbl" />

<TextView
   android:id="@+id/FNameLbl"
   android:layout_width="wrap_content"
   android:layout_height="wrap_content"
   android:text="Father Name"
   android:textStyle="bold"
   app:layout_constraintBottom_toTopOf="@+id/GenderLbl"
   app:layout_constraintEnd_toStartOf="@+id/FNameBox"
   app:layout_constraintStart_toStartOf="parent"
   app:layout_constraintTop_toBottomOf="@+id/nameLbl" />

<RadioGroup
   android:id="@+id/GenderGroup"
   android:layout_width="wrap_content"
   android:layout_height="wrap_content"
   app:layout_constraintStart_toEndOf="@+id/GenderLbl"
   app:layout_constraintBottom_toTopOf="@+id/PlaceBox"
   app:layout_constraintTop_toBottomOf="@+id/FNameBox"
   app:layout_constraintEnd_toEndOf="parent">

   <RadioButton
      android:id="@+id/MaleBtn"
      android:layout_width="wrap_content"
      android:layout_height="wrap_content"
      android:text="Male" />

   <RadioButton
```

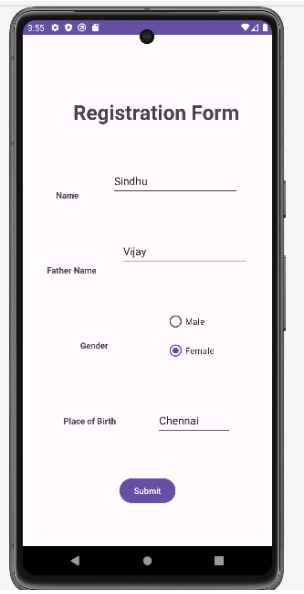
```
        android:id="@+id/FemaleBtn"
        android:layout_width="wrap_content"
        android:layout_height="wrap_content"
        android:text="Female" />

    </RadioGroup>

    <EditText
      android:id="@+id/NameBox"
      android:layout_width="wrap_content"
      android:layout_height="wrap_content"
      android:ems="10"
      android:hint="Enter Name"
      android:inputType="text"
      android:minHeight="48dp"
      app:layout_constraintBottom_toTopOf="@+id/FNameBox"
      app:layout_constraintEnd_toEndOf="parent"
      app:layout_constraintStart_toEndOf="@+id/nameLbl"
      app:layout_constraintTop_toBottomOf="@+id/header" />

    <EditText
      android:id="@+id/FNameBox"
      android:layout_width="wrap_content"
      android:layout_height="wrap_content"
      android:ems="10"
      android:hint="Enter Father Name"
      android:inputType="text"
      android:minHeight="48dp"
      app:layout_constraintBottom_toTopOf="@+id/GenderGroup"
      app:layout_constraintEnd_toEndOf="parent"
      app:layout_constraintStart_toEndOf="@+id/FNameLbl"
      app:layout_constraintTop_toBottomOf="@+id/NameBox" />

    <Button
      android:id="@+id/SubmitBtn"
      android:layout_width="wrap_content"
      android:layout_height="wrap_content"
      android:text="Submit"
      app:layout_constraintEnd_toEndOf="parent"
      app:layout_constraintStart_toStartOf="parent"
      app:layout_constraintTop_toBottomOf="@+id/PlaceBox"
      app:layout_constraintBottom_toBottomOf="parent"/>


</androidx.constraintlayout.widget.ConstraintLayout>
```

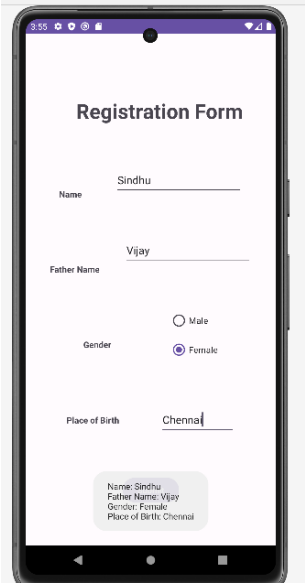
MainActivity.java:

```
package com.example.registrationformacrg;

import androidx.appcompat.app.AppCompatActivity;
```

```
import android.annotation.SuppressLint;
import android.os.Bundle;
import android.view.View;
import android.widget.ArrayAdapter;
import android.widget.AutoCompleteTextView;
import android.widget.Button;
import android.widget.EditText;
import android.widget.RadioButton;
import android.widget.RadioGroup;
import android.widget.TextView;
import android.widget.Toast;

public class MainActivity extends AppCompatActivity {
    EditText name,fname;
    AutoCompleteTextView places;
    RadioGroup rg;
    RadioButton rb;
    Button btn;
    String[] Locations = {"Bangalore","Delhi","Pune","Bombay","Chennai","Hyderabad","Calcutta"};
    @SuppressLint("MissingInflatedId")
    @Override
    protected void onCreate(Bundle savedInstanceState) {
        super.onCreate(savedInstanceState);
        setContentView(R.layout.activity_main);
        name = findViewById(R.id.NameBox);
        fname = findViewById(R.id.FNameBox);
        places = findViewById(R.id.PlaceBox);
        btn = findViewById(R.id.SubmitBtn);
        ArrayAdapter adapter = new ArrayAdapter(this, android.R.layout.select_dialog_item,Locations);
        places.setThreshold(1);
        places.setAdapter(adapter);
        rg = findViewById(R.id.GenderGroup);
        rg.setOnCheckedChangeListener(new RadioGroup.OnCheckedChangeListener() {
            @Override
            public void onCheckedChanged(RadioGroup group, int checkedId) {
                rb = findViewById(checkedId);
            }
        });
        btn.setOnClickListener(new View.OnClickListener() {
            @Override
            public void onClick(View v) {
                String res1 = "Name: "+name.getText().toString()+"\nFather Name:
"+fname.getText().toString()+"\nGender: "+rb.getText()+"\nPlace of Birth:
"+places.getText().toString();
                Toast.makeText(MainActivity.this,res1,Toast.LENGTH_SHORT).show();
            }
        });
    }
```

Output:


Registration Form
Name
Sindhu
Father Name
Vijay
Gender
Male
Female
Place of Birth
Chennai
Submit


Registration Form
Name
Sindhu
Father Name
Vijay
Gender
Male
Female
Place of Birth
Chennai
Name: Sindhu
Father Name: Vijay
Gender: Female
Place of Birth: Chennai

Experiment – 4

Aim: Design an application to fill the course selection of discipline electives for upcoming semester using Spinner by reading the input from user

Solution:

activity_main.xml:

```
<?xml version="1.0" encoding="utf-8"?>
<androidx.constraintlayout.widget.ConstraintLayout
xmlns:android="http://schemas.android.com/apk/res/android"
   xmlns:app="http://schemas.android.com/apk/res-auto"
   xmlns:tools="http://schemas.android.com/tools"
   android:layout_width="match_parent"
   android:layout_height="match_parent"
   tools:context=".MainActivity">

   <TextView
      android:id="@+id/header"
      android:layout_width="wrap_content"
      android:layout_height="wrap_content"
      android:text="Electives Selection"
      android:textSize="34sp"
      android:textStyle="bold"
      app:layout_constraintBottom_toTopOf="@+id/nameBox"
      app:layout_constraintEnd_toEndOf="parent"
      app:layout_constraintStart_toStartOf="parent"
      app:layout_constraintTop_toTopOf="parent" />

   <TextView
      android:id="@+id/nameLbl"
      android:layout_width="wrap_content"
      android:layout_height="wrap_content"
      android:text="Student Name"
      android:textStyle="bold"
      app:layout_constraintStart_toStartOf="parent"
      app:layout_constraintEnd_toStartOf="@+id/nameBox"
      app:layout_constraintTop_toBottomOf="@+id/header"
      app:layout_constraintBottom_toTopOf="@+id/rollnoLbl"/>

   <EditText
      android:id="@+id/nameBox"
      android:layout_width="wrap_content"
      android:layout_height="wrap_content"
      android:ems="10"
      android:minHeight="48dp"
      android:hint="Enter Student Name"
      android:inputType="text"
      app:layout_constraintStart_toEndOf="@+id/nameLbl"
      app:layout_constraintTop_toBottomOf="@+id/header"
      app:layout_constraintBottom_toTopOf="@+id/rollnoBox"
```

```
    app:layout_constraintEnd_toEndOf="parent"/>

<TextView
    android:id="@+id/rollnoLbl"
    android:layout_width="wrap_content"
    android:layout_height="wrap_content"
    android:text="Roll No"
    android:textStyle="bold"
    app:layout_constraintStart_toStartOf="parent"
    app:layout_constraintTop_toBottomOf="@+id/nameLbl"
    app:layout_constraintEnd_toStartOf="@+id/rollnoBox"
    app:layout_constraintBottom_toTopOf="@+id/delbl"/>

<EditText
    android:id="@+id/rollnoBox"
    android:layout_width="wrap_content"
    android:layout_height="wrap_content"
    android:ems="10"
    android:minHeight="48dp"
    android:hint="Enter Student RollNo"
    android:inputType="text"
    app:layout_constraintStart_toEndOf="@+id/rollnoLbl"
    app:layout_constraintEnd_toEndOf="parent"
    app:layout_constraintBottom_toTopOf="@+id/delbl"
    app:layout_constraintTop_toBottomOf="@+id/nameBox"/>

<TextView
    android:id="@+id/delbl"
    android:layout_width="wrap_content"
    android:layout_height="wrap_content"
    android:text="Discipline Elective"
    android:textStyle="bold"
    app:layout_constraintBottom_toTopOf="@+id/electivesBox"
    app:layout_constraintEnd_toEndOf="parent"
    app:layout_constraintHorizontal_bias="0.054"
    app:layout_constraintStart_toStartOf="parent"
    app:layout_constraintTop_toBottomOf="@+id/rollnoLbl" />

<Spinner
    android:id="@+id/electivesBox"
    android:layout_width="match_parent"
    android:layout_height="wrap_content"
    android:minHeight="48dp"
    app:layout_constraintBottom_toTopOf="@+id/SubmitBtn"
    app:layout_constraintEnd_toEndOf="parent"
    app:layout_constraintStart_toStartOf="parent"
    app:layout_constraintTop_toBottomOf="@id/delbl"/>

<Button
    android:id="@+id/SubmitBtn"
```

```
        android:layout_width="wrap_content"
        android:layout_height="wrap_content"
        android:text="Submit"
        android:onClick="displayDetails"
        app:layout_constraintTop_toBottomOf="@+id/electivesBox"
        app:layout_constraintStart_toStartOf="parent"
        app:layout_constraintBottom_toTopOf="@+id/result"
        app:layout_constraintEnd_toStartOf="@+id/ResetBtn"/>

    <Button
        android:id="@+id/ResetBtn"
        android:layout_width="wrap_content"
        android:layout_height="wrap_content"
        android:text="Reset"
        android:onClick="resetDetails"
        app:layout_constraintStart_toEndOf="@+id/SubmitBtn"
        app:layout_constraintEnd_toEndOf="parent"
        app:layout_constraintTop_toBottomOf="@+id/electivesBox"
        app:layout_constraintBottom_toTopOf="@+id/result"/>

    <TextView
        android:id="@+id/result"
        android:layout_width="wrap_content"
        android:layout_height="wrap_content"
        android:gravity="center_horizontal"
        app:layout_constraintEnd_toEndOf="parent"
        app:layout_constraintStart_toStartOf="parent"
        app:layout_constraintBottom_toBottomOf="parent"
        app:layout_constraintTop_toBottomOf="@+id/SubmitBtn"/>
</androidx.constraintlayout.widget.ConstraintLayout>
```

strings.xml:

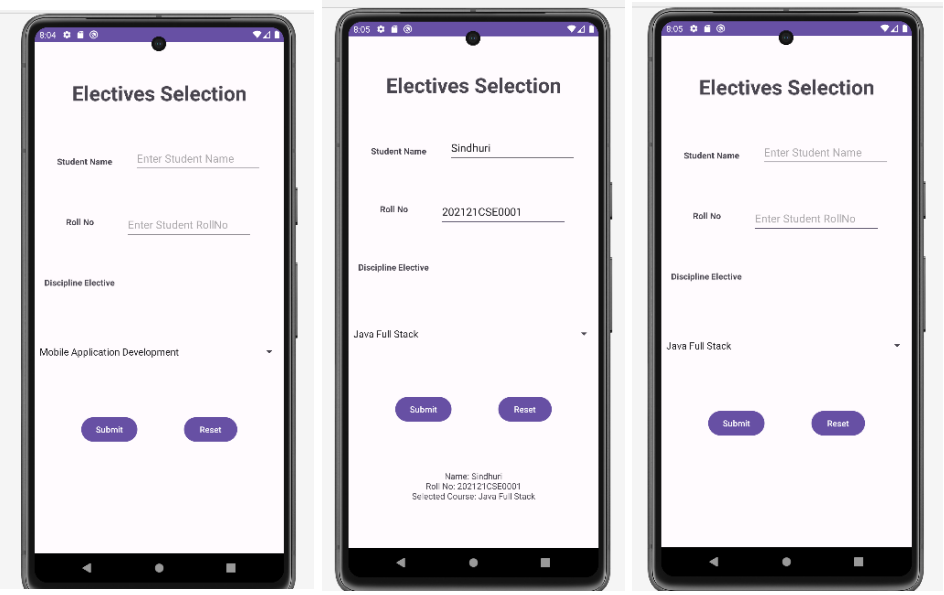
```
<resources>
    <string name="app_name">DESelectionSpinner</string>
    <string-array name="courses">
        <item>Mobile Application Development</item>
        <item>Java Full Stack</item>
        <item>.Net Full Stack</item>
    </string-array>
</resources>
```

MainActivity.java:

```
package com.example.deselectionspinner;

import androidx.appcompat.app.AppCompatActivity;

import android.os.Bundle;
import android.view.View;
```

```
import android.widget.AdapterView;
import android.widget.ArrayAdapter;
import android.widget.Button;
import android.widget.EditText;
import android.widget.Spinner;
import android.widget.TextView;

public class MainActivity extends AppCompatActivity {
   EditText name,rollno;
   Spinner elec;
   TextView res;
   String selectedCourse;
   @Override
   protected void onCreate(Bundle savedInstanceState) {
      super.onCreate(savedInstanceState);
      setContentView(R.layout.activity_main);
      name = findViewById(R.id.nameBox);
      rollno = findViewById(R.id.rollnoBox);
      elec = findViewById(R.id.electivesBox);
      res = findViewById(R.id.result);
      ArrayAdapter adapter = new ArrayAdapter(this,
android.R.layout.simple_spinner_item,getResources().getStringArray(R.array.courses));
      adapter.setDropDownViewResource(android.R.layout.simple_spinner_dropdown_item);
      elec.setAdapter(adapter);
      elec.setOnItemSelectedListener(new AdapterView.OnItemSelectedListener() {
         @Override
         public void onItemSelected(AdapterView<?> parent, View view, int position, long id) {
            selectedCourse = parent.getItemAtPosition(position).toString();
         }

         @Override
         public void onNothingSelected(AdapterView<?> parent) {

         }
      });
   }
   public void displayDetails(View v) {
      res.setText("Name: "+name.getText().toString()+"\nRoll No:
"+rollno.getText().toString()+"\nSelected Course: "+selectedCourse);
   }
   public void resetDetails(View v){
      name.setText("");
      rollno.setText("");
      res.setText("");
   }
}
```

Output:


Electives Selection
Student Name
Enter Student Name
Roll No
Enter Student RollNo
Discipline Elective
Mobile Application Development
Submit
Reset
Electives Selection
Student Name
Sindhuri
Roll No
202121CSE0001
Discipline Elective
Java Full Stack
Submit
Reset
Name: Sindhuri
Roll No: 202121CSE0001
Selected Course: Java Full Stack
Electives Selection
Student Name
Enter Student Name
Roll No
Enter Student RollNo
Discipline Elective
Java Full Stack
Submit
Reset